Bruno Tavares Meixedo

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Education

University of Aveiro — **Aveiro, Portugal**
Master's in Informatics Engineering — current average 16.3/20 — *Sep 2025 – Present*

University of Aveiro — **Aveiro, Portugal**
Bachelor's in Informatics Engineering — *Sep 2021 – Jun 2025*

Research Experience

Master's Thesis – Reading the Room — **Sep 2026 – Present**
DETI / IEETA – University of Aveiro — *Aveiro*

- Investigating whether smart environments can interact better when they know who is in a room, where they are and what they are facing
- Designing a device-free pipeline that combines RGB-D camera data (body pose, head orientation, gestures) with 3D object positions into a shared spatial and temporal context model
- Handling real-world conditions: multiple occupants, occlusion, distance and lighting
- Exposing this context to multimodal interaction systems and comparing interaction with and without it

Student Researcher – Agenda ILLIANCE — **Jan 2026 – Jun 2026**
IEETA – University of Aveiro — *Aveiro*

- Developed a low-cost eye-tracking platform that estimates where people look from a standard webcam, with no dedicated hardware (MediaPipe, MGazeNet, per-user SVR calibration)
- Supported visual behaviour analysis in four contexts: screen interfaces and videos, with zero-shot detection of areas of interest (YOLO-World, GroundingDINO, ByteTrack); physical surfaces such as paper prototypes, mapped onto a digital twin; and objects around the room, identified from head pose (solvePnP)
- Processed gaze in a Medallion pipeline that classifies fixations (I-VT) and computes attention metrics per area (dwell time, TTFF, revisits), shown in a Streamlit dashboard with heatmaps and scan paths

Projects

Open Data Portal | *Spring Boot, React, FastAPI, SQL Server, Parquet*

- Built a port data analytics portal with W3C PROV-based provenance tracking and data origin graphs
- Added data quality checks and ML-based outlier detection as Python microservices

E-Procurement Data Lakehouse | *Python, Parquet, Dremio, Selenium, Streamlit*

- Built a Medallion lakehouse for Portuguese and EU procurement data (BASE, EU Tenders, VORTAL)
- Implemented RapidFuzz entity resolution, quality scoring and Gemini-based text-to-SQL on Dremio

Vehicle & Parking Management System | *Spring Boot, React, Kafka, Redis, Azure*

- Built parking management microservices with Keycloak (OAuth2), Kafka and Redis behind Nginx
- Deployed on Azure with Terraform and GitHub Actions; tested with Cucumber and k6; monitored with Prometheus, Grafana and ELK

Skills

Languages & Tools: Python, Java, TypeScript, JavaScript, C++, C, SQL, MATLAB, Git, Linux, QGIS
Vision & Data: MediaPipe, OpenCV, YOLO-World, GroundingDINO, OWLv2, Pandas, scikit-learn, Airflow
Systems & Web: React, Spring Boot, FastAPI, PostgreSQL, Redis, Kafka, Docker, Terraform, CUDA, MPI

Additional

Certificate of Pedagogical Competencies (CCP), IEFP, 2026 | Languages: Portuguese (native), English